



DACL+: domain-adapted contrastive learning for enhanced low-resource language representations in document clustering tasks

Dimitrios Zaikis¹ · Ioannis Vlahavas¹

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Abstract

Low-resource languages in natural language processing present unique challenges, marked by limited linguistic resources and sparse data. These challenges extend to document clustering tasks, where the need for meaningful and semantically rich representations is crucial. Along with the emergence of transformer-based language models (LM), the need for vast amounts of training data has also increased significantly. To this end, we introduce a domain-adapted contrastive learning approach for low-resource Greek document clustering. We introduce manually annotated datasets, essential for LM pre-training and clustering tasks, and extend the investigations by combining Greek BERT and Longformer models. We explore the efficacy of various domain adaptation pre-training objectives and of further pre-training the LMs using contrastive learning with diverse loss functions on datasets generated from a classification corpus. By maximizing the similarity between positive examples and minimizing the similarity between negative examples, our proposed approach learns meaningful representations that capture the underlying structure of the documents. We demonstrate that our proposed approach significantly improves the accuracy of clustering tasks, with an average improvement of up to 50% compared to the base LM, leading to enhanced performance in unsupervised learning tasks. Furthermore, we show how combining language models optimized for different sequence lengths improves performance and compare this approach against an unsupervised graph-based summarization method. Our findings underscore the importance of effective document representations in enhancing the accuracy of clustering tasks in low-resource language settings.

Keywords Representation Learning · Clustering · Low-Resource Language · Contrastive Learning · Domain Adaptation · NLP

1 Introduction

Natural language processing (NLP) is a rapidly growing field of study that focuses on developing techniques and algorithms to enable computers to understand and interpret human language. With the increasing amount of unstructured data in the form of text, the ability to analyze and extract insights from such data has become increasingly

important. In this context, generating document representations is a critical step in converting raw text data into numerical vectors that can be used for various text analysis tasks, such as document clustering [1]. This is especially important for low-resource languages, where it becomes more challenging to develop effective NLP models due to the lack of available data and resources [2].

Document clustering aims to automatically organize documents into clusters based on their similarity to documents in other clusters. This approach traditionally required the use of a feature matrix, such as a tf-idf matrix, to describe the corpus, followed by a clustering algorithm applied to the matrix. However, more recent approaches using neural word embeddings have gained popularity due

✉ Dimitrios Zaikis
dimitriz@csd.auth.gr

Ioannis Vlahavas
vlavavas@csd.auth.gr

¹ School of Informatics, Aristotle University of Thessaloniki, Thessaloniki, Greece

to their ability to yield dense semantic representations that significantly improve the models' performances [3].

Similarly, topic modeling is a document clustering-based task that involves discovering patterns of word usage in documents using various methodologies, where documents are modeled as a mixture of topics represented as a distribution of words [4]. Similar to document clustering, it can be used for soft partition clustering, assigning a probability distribution over a set of topics to each document, and allowing for a probabilistic degree of affiliation with each cluster. Furthermore, the topic representation contains the word distribution for each topic, making it easier to interpret. [5].

While representation learning in English has become a very active study area in recent years, there has been relatively little published work for low-resource languages such as Greek [6, 7]. Due to its rich morphological properties, written Greek is a particularly difficult language for NLP in general and for document clustering in particular (high inflection, stressing rules, etc.). This makes it challenging to create effective models for the various NLP tasks in Greek, and highlights the need for further research in this area to improve the accuracy and reliability of such systems.

In this paper, we build upon our prior work [8], where we proposed a domain-adapted contrastive learning approach for low-resource language representations in document clustering tasks for the Greek language. To facilitate this research, we also introduce and describe newly collected and manually annotated datasets for language model second-phase pre-training and clustering tasks representation learning. We introduce and detail newly collected, manually annotated datasets crucial for language models (LM) second-phase pre-training and clustering tasks representation learning. We extend the investigation on the effectiveness of combining domain-adapted Greek BERT and Longformer models using contrastive learning with different loss functions that are used to fine-tune the models on training data across three source domains for document clustering tasks. Finally, we highlight the importance of adapting LMs to specific domains and the importance of further training for improved representation generation.

The main contributions of our work are:

- A novel approach to representation learning using domain adaptation on pre-trained language models.
- Document-level clustering using a combination of a BERT-based LM with a Greek pre-trained Longformer LM.
- A novel training approach that can be applied to other high and low-resource languages.

2 Related work

Effective semantic representations for documents are integral to the field of NLP and require vast amounts of training data. To overcome this challenge, a recent study [9] introduced a method grounded in implicit maximum likelihood estimation (IMLE), where a generative neural network is trained on random noise to produce synthetic data closely resembling the original dataset. This approach effectively merges the generative process with a clustering procedure, yielding promising results across diverse datasets of varying complexities. Similarly, the work in [10] builds on the foundations of ClusterGAN [11] and IMLE, presenting the neural implicit maximum likelihood clustering (NIMLC) method. NIMLC addresses deficiencies inherited from generative adversarial networks, such as mode collapse and training instability, by integrating ClusterGAN concepts into the IMLE framework. Notably, NIMLC demonstrates improved performance on both synthetic and real datasets, as evidenced by experimental comparisons against various clustering approaches. These comparisons underscore the potential of NIMLC, particularly in effectively clustering both cloud-shaped and ring-shaped synthetic data using consistent hyperparameter settings.

Recently, distributed document representations have gained importance due to their accuracy as semantic text representations. However, the context of a document differs significantly between different domains [12]. Deep neural networks are capable of learning transferable features, but recent research shows that the transferability of features decreases significantly in higher task-specific layers as the domain discrepancy increases. This means that the characteristics in deeper layers of the networks largely depend on specific datasets and tasks, which cannot be reliably transferred to new tasks [13]. To address this issue, adaptation approaches have become popular, which aim to reduce domain discrepancy via an adversarial objective with respect to a domain classifier. Ultimately, these approaches aim to improve the model's accuracy in different domains [14].

Moreover, in some scenarios where learning tasks vary across domains, adaptation approaches have become popular strategies to reduce domain discrepancy via adversarial objectives with respect to a domain classifier [14]. However, the effectiveness of these approaches diminishes as domain discrepancy increases. Recent research, such as that by Faralli et al. [15], proposes domain-driven word sense disambiguation (WSD) methods that construct glossaries for multiple domains using pattern-based bootstrapping techniques. Although their work primarily focuses on WSD, it underscores the importance of evaluating domain-

specific text senses, contributing valuable insights into the domain adaptation process.

Neelakantan et al. [16] proposed a method that uses WSD and word embedding learning simultaneously, thereby learning several embeddings for each word type. However, their proposed methodologies are limited to a single domain and do not take into account how representations differ between domains.

Transformers have significantly enhanced document representation learning, achieving state-of-the-art results in various NLP tasks. This success can be attributed, in part, to their self-attention mechanism, which enables the network to capture contextual information throughout the entire sequence. Two self-attention methods have been explored. The first is a left-to-right approach that processes the sentence in left-to-right chunks that can be effective for autoregressive language modeling but is unsuitable for representation learning tasks that require bidirectional context [17]. BPTransformer [18] investigated the machine translation task, but did not explore the pre-training and fine-tuning setting. Meanwhile, a more general approach involves defining sparse attention patterns that avoid computing the entire quadratic attention matrix multiplication. Transformers have significantly enhanced document representation learning, with approaches such as sparse transformer [19] offering novel ways to generate sparse representations using dilated sliding window blocks.

Since documents usually have a large body of text that contains thousands of tokens, several task-specific strategies have been devised to work around the 512 tokens limitation of pre-trained LMs. One straightforward approach is to abbreviate the documents [20]. Another option is to split the text into 512-byte chunks, and process each chunk individually before combining the results with a task-specific model [21]. For multi-hop and open domain QA tasks, a two-stage architecture is a popular solution, in which the first stage retrieves relevant documents that are then passed on to the second stage for response extraction [22]. However, all of these methods suffer from information loss or cascading errors, as a result of truncation or two-stage processing [23].

Recently, dense representation generation models have become more prevalent in document retrieval problems, outperforming traditional sparse vector space models. These models typically use a Bi-encoder structure to achieve high efficiency. For the document clustering task, recent approaches frequently utilize bidirectional encoder representations from transformers (BERT) based models [24]. Sentence BERT (SBERT) [25], which is a variation of the conventional BERT network, generates semantically meaningful phrase embeddings using Siamese and triplet network architectures, leading to improved clustering

performance compared to traditional sparse vector space models.

In the Greek document clustering domain, implementations rely on FastText, a Word2vec-like model, to create clusters from social media texts [26]. Furthermore, leveraging various representation learning techniques using transformer-based LM, such as domain adaptation and contrastive learning, has proved to improve language understanding in a multistage language modeling approach for Greek classification tasks [27].

3 Materials and methods

In this section, we present a detailed description of the datasets, the LM domain adaptation and the contrastive learning approach of our proposed methodology. Specifically, we created three datasets for second-phase pre-training, contrastive representation learning and the clustering tasks, correspondingly and provide an overview of the data collection and annotation process. Furthermore, we provide an in-depth overview of our proposed approach to representation learning with domain-adapted contrastive learning for clustering tasks. Figure 1 illustrates the flowchart of our proposed approach.

3.1 Datasets

Due to the lack of necessary resources, including datasets and tools, the Greek language is considered a low-resource language (LRL), emphasizing the necessity for large amounts of training data. To capture target-specific and period-specific characteristics in our representation learning approach, both unsupervised and supervised learning phases are necessary, requiring both unlabeled domain-specific data and labeled target-specific data. Our proposed approach utilizes transformer-based LMs for domain adaptation (second-phase pre-training) in the unsupervised phase and extends the training with contrastive learning in the supervised phase. To allow potential generalizability, we focused on collecting data from three distinct sources: internet, social media and press. The internet data consist of a collection of articles from blog posts, news sites and news aggregators. The social media data are comprised of posts from Facebook, Twitter and Instagram, while the press data include text clips from published newspapers and newsletters that are identical to the printed copies. It is important to note that the collected data did not include any personal information or sensitive data, such as user handles, to ensure privacy and ethical considerations.

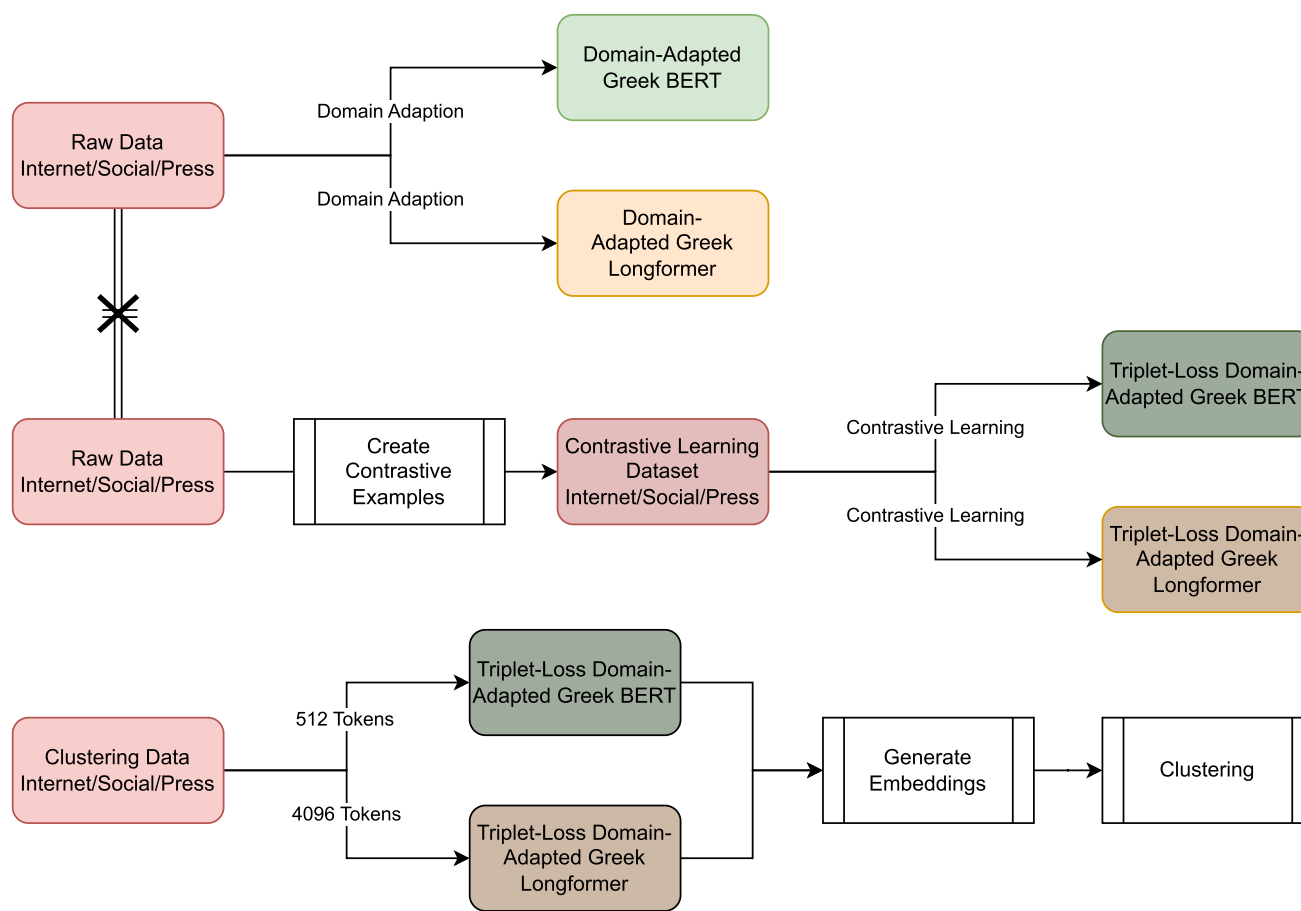


Fig. 1 Flowchart of our proposed approach to domain-adapted contrastive learning for clustering tasks. Different datasets are used for domain adaptation and the creation of the contrastive learning datasets

3.1.1 Language model domain adaptation dataset

The domain adaptation dataset was created to aid in the second-phase pre-training of the Greek BERT model [28] and the pre-training of a randomly initialized Longformer [17] model. Pre-training language models on domain-specific data have proved to be effective in improving their performance on tasks within that domain, since even high-parameters models can struggle to capture the complexity of a single domain [29].

The dataset consists of sentence-level texts gathered from the aforementioned online sources and domains, but the sentences come from different contexts and are much larger in volume. In total, there are 1,590,409 instances, split into 25,118,855 sentences from all categories, with an average length of 24.2 tokens and a high variance of ± 28.5 . The probability of a word being masked is 15%, and the dataset was split for training and testing with a ratio of 9:1. Table 1 provides detailed information about the source types and the instances in each category.

3.1.2 Contrastive learning dataset

We initially collected textual data using specific keywords (i.e., car brands, cosmetic brands, etc.) to create a pool with 13,019 instances from internet sources, 12,391 instances from social media sources and 23,089 instances from press sources. The content was collected by randomly selecting articles, posts and text to ensure a uniform distribution among each source, user and topic. Additionally, the type of content such as “news article,” “editorial,” “comment” and “post” was automatically inferred and annotated. Non-Greek content was excluded, and the focus was on achieving diversity of content for each class.

Our aim was to gather representative samples of articles, posts and text clips from each source, rather than selecting content based on any specific criteria. Finally, three expert annotators manually annotated the examples to determine whether the text was directly, indirectly or not related to the accompanying keyword (focus) based on a set of pre-determined guidelines that were adjusted during the annotation process. The annotation was carried in batches

Table 1 Number of instances by category and domain in the LM domain adaptation dataset

Domain	Category	Instances	Ratio (%)	Total
Internet	Blogs	26,507	15.11	1,205,321
	News sites/Aggregators	1,178,814	74.02	
Social Media	Facebook	144,813	19.38	308,141
	Twitter	144,530	9.08	
	Instagram	18,798	1.18	
	Free press	5,212	3.85	
Press	Sports	1,930	0.12	76,947
	Economics	38,656	2.43	
	Local	31,149	1.96	

and evaluated with Cohen’s Kappa [30] and achieved a score of 0.84, indicating a very high agreement rate.

The contrastive learning dataset is comprised of sentence triplets, the anchor (S^A), a positive (S^+) and a negative (S^-) sentence, inspired by the approach presented in [31]. Initially, a random sentence is selected as anchor. The positive example is selected based on the same keyword as the anchor in addition to having the same focus. Since the selection of the negative examples is crucial, we also considered two sets of negatives for each anchor and positive pair. The first negative instance is randomly chosen from the corpus but has a completely different keyword (K), referred to as “easy negative (S_E^-),” while the second set is selected based on the same keyword but different focus (F), referred to as “hard negative (S_H^-).” Equations 1 and 2 describe the triplets selection process:

$$S_E^-(S^A, S^+) = s | s \notin \{S^A, S^+\} \wedge K_s \neq K_{S^A} \quad (1)$$

$$S_H^-(S^A, S^+) = s | s \notin \{S^A, S^+\} \wedge K_s = K_{S^A} \wedge F_s \neq F_{S^A} \quad (2)$$

where the symbols $|$, $\notin$ and $\wedge$ mean “such that,” “not an element of” and “and,” respectively, and s denotes the candidate sentence. For each anchor and positive pair S^A, S^P we pick two S_E^- and three S_H^- instances.

The intuition behind using hard negatives is that the model should be able to distinguish between directly related instances and unrelated instances sampled randomly from the entire corpus when given a query sentence. However, randomly selected negative examples may be too easy for the model to distinguish from the positive examples.

The final datasets consist of 55,170 instances from internet sources, 99,875 instances from social media sources and 75,485 instances from press sources resulting in a total of 230,530 instances from all the sources combined, as shown in Table 2. The dataset was split in a stratified manner, with 80% used for training, 10% for validation and 10% for testing.

3.1.3 Clustering dataset

Similar to the previous datasets, the clustering dataset is comprised of texts from internet, social media and press sources collected using a wide variety of keywords with a total of 283,052 instances. Each instance has the related keyword, the text that can vary greatly in length and the title where applicable as social media posts usually do not have one. Finally, the total number of unique keywords used across all sources is 3,393 from multiple categories such as heavy industries, retail, government, contractors, pharmaceuticals and much more.

3.2 Domain-adapted contrastive representation learning

In our proposed contrastive representation learning approach, we leverage two transformer-based LMs as foundational components: Greek BERT and the Longformer [17, 28]. These models serve as key building blocks in enhancing document clustering in the low-resource language, Greek. The Greek BERT model is pre-trained on a large corpus of Greek texts, providing a robust foundation for understanding the Greek language [28]. Simultaneously, the Longformer LM, introduced in its preliminary version in [8] and fully pre-trained in its final iteration in [27], further contributes to our framework.

The transformer architecture is composed of two main components: the encoder and the decoder. The encoder processes the input sequence and produces a sequence of contextualized embeddings, while the decoder uses those embeddings to generate an output sequence. In the case of BERT and Longformer, they only use the encoder part of the transformer. By using multiple stacked layers of the transformer encoder, both models can take advantage of the self-attention mechanism [32], which allows the model to attend to different parts of the input sequence and capture complex contextual relationships between tokens.

One key difference between them is in their ability to handle long sequences of text. BERT was designed to

Table 2 Summary of the contrastive learning dataset statistics

Source	Initial number of instances	Number of contrastive triplets
Internet	1,205,321	55,170
Social Media	308,141	99,875
Press	76,947	75,485
Total	1,590,409	230,530

process relatively short input sequences of up to 512 tokens, which can be a limitation when working with longer texts such as long documents or articles, while Longformer is able to handle longer input sequences of up to several thousands of tokens.

BERT employs the traditional self-attention mechanism where the attention is computed by considering all positions (or tokens) in the input sequence simultaneously. Each position attends to every other position, capturing bidirectional dependencies within the context. This enables BERT to capture the relationships between words and their contextual dependencies, making it particularly effective in downstream tasks.

On the other hand, the Longformer introduces a novel attention mechanism designed to handle long sequences more efficiently. In traditional transformers, the self-attention mechanism has a quadratic complexity with respect to sequence length, making it computationally expensive for lengthy documents. The Longformer addresses this limitation by introducing a combination of global and local attention mechanisms. While maintaining a global view of the sequence, the Longformer also focuses on a subset of tokens within a certain window, allowing it to scale efficiently to longer documents.

The global attention ensures that the Longformer can capture broad contextual information, while the local attention mechanism facilitates the modeling of fine-grained relationships within shorter spans of the document. This dual attention strategy makes the Longformer well-suited for tasks involving extensive documents, such as document clustering, where a balance between global and local context is crucial.

Both models utilize the masked language modeling (MLM) pre-training objective where random tokens in the input sequence are masked and the models are trained to predict the masked tokens.

3.2.1 Domain adaptation

Domain adaptation is the process of adapting a pre-trained language model to a specific domain or topic by fine-tuning it on domain-specific data. This can be especially useful for low-resource LMs that do not perform well on text that is related to a specific topic or domain, such as medical texts, legal documents or social media posts.

To perform the domain adaptation, also referred to as LM alignment, we fine-tuned the pre-trained Greek BERT LM on the domain adaptation dataset presented in Sect. 3.1.1 using the MLM training objective and using transformer-based sequential denoising auto-encoder (TSDAE) [33], as shown in Fig. 2.

These forms of second-phase pre-training involve retraining the model on the new data, while keeping the pre-trained weights from the original LM, effectively updating the model's parameters to better fit the new domain knowledge, allowing it to capture the specific language patterns and structures. For the Greek Longformer, we first trained a randomly initialized model from scratch using the same domain adaptation dataset. In the second iteration, we utilized the fully pre-trained Greek Longformer introduced in [27] and aligned it by leveraging the domain adaptation dataset using both pre-training task.

MLM (Fig. 2a) is a type of unsupervised learning where the model is trained to predict missing or masked words in a sentence. The objective is to learn a generalized representation of language that captures contextual information and semantic relationships. The process involves randomly masking certain words in a given sentence (15% of words) by replacing them with a special [MASK] token. Then model is then trained to predict those masked words based on the surrounding context, helping the model understand the relationships between words and the overall structure of language.

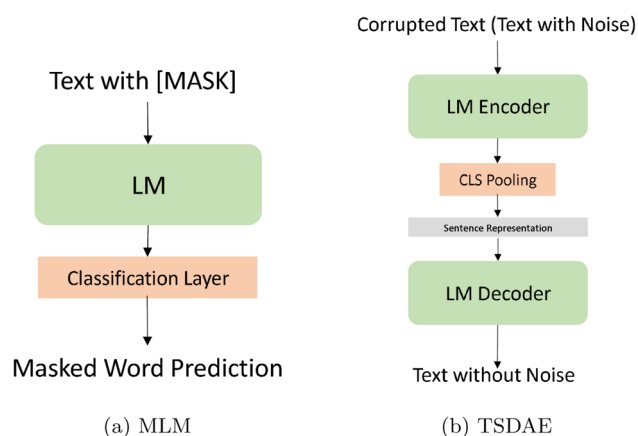


Fig. 2 Domain adaptation (LM alignment) approaches using **a** masked language modeling and **b** transformer-based sequential denoising auto-encoder

TSDAE (Fig. 2b) is an unsupervised sentence embedding method that uses a modified encoder-decoder transformer design where the key and the value of the cross-attention are confined to the sentence embedding. During preprocessing the unlabeled input sentences are corrupted by deleting 60% of their content to introduce input noise. To achieve high-quality reconstruction, it is essential for the sentence embedding generated by the encoder to effectively capture semantic information. This is accomplished by employing the pre-trained transformer models, as the encoder. In this setup, the weights of the decoder are initialized by copying them from the encoder. This process leverages the pre-trained knowledge encoded in the encoder to guide the generation of meaningful and contextually rich sentence representations during the decoding phase.

3.2.2 Contrastive learning

By implementing contrastive learning with the use of various adversarial loss functions [34] as a type of contrastive loss, we further train the LMs while leveraging the label information more effectively. The aim is to pull instances that belong to the same class in the embedding space closer, while simultaneously pushing instances from different classes further apart.

Document-level contrastive learning aims to learn representations of documents that capture semantic and contextual information, while also capturing the differences between documents, effectively learning better representations that can be used for clustering-based downstream tasks. The LMs are trained to differentiate between the examples by maximizing the similarity between the representations of positive examples and minimizing the similarity between the representations of negative examples, by encoding the different documents and output embeddings for each document, which can be compared using a contrastive loss function.

Triplet loss (TL), contrastive loss (CL) and Cosine similarity loss (CSL) were used to further pre-train and fine-tune the LMs on the training data. Given a triplet of (anchor, positive, negative), TL minimizes the distance between anchor and positive while it maximizes the distance between anchor and negative, using the loss function as described in Eq. 3.

$$\mathcal{L}_{TL} = \max(|\text{anchor} - \text{positive}| - |\text{anchor} - \text{negative}| + \text{margin}, 0) \quad (3)$$

CL (Eq. 4) changes the distance between two embeddings based on a label of either 0 or 1, where 1 denotes similar pairs where the distance will be reduced, and 0 denotes dissimilar pairs where the distance of the embeddings should be increased.

$$\mathcal{L}_{CL} = 0.5(\text{labels} \cdot \text{distances}^2 + (1 - \text{labels}) \cdot (\max(0, \text{margin} - \text{distances})^2)) \quad (4)$$

Where *distances* denotes the Euclidean distances between the embeddings, *margin* the margin parameter used in the loss calculation and $\max(0, \text{margin} - \text{distances})$ represents the ReLU function, ensuring that negative values are replaced by zeros. The CL loss dataset was created using the TL dataset as a reference, assigning the label 1 to anchor-positive pairs and the label 0 to anchor-negative pairs.

CSL (Eq. 5) allows the network to recognize the similarity of sentences by measuring the Cosine similarity between two embeddings generated by the same model, $u = \text{model}(\text{sentence}_A)$ and $v = \text{model}(\text{sentence}_B)$, and compares the result to the gold similarity score, similar to CL.

$$\mathcal{L}_{CSL} = \|\text{input_label} - \cos_score_transformation(\cos(u, v))\|_2 \quad (5)$$

Where *input_label* denotes the gold similarity score, $\cos(u, v)$ the Cosine similarity between vectors *u* and *v* and *cos_score_transformation* represents the transformation functions that is applied on top of Cosine similarity, which in this case is the identify function that does not change the result. $\|\cdot\|_2$ denotes the L2 (Euclidean) norm. Similarly to the CL loss dataset, the CSL loss dataset was created using the TL loss dataset as a reference with the distinction being that training examples now include non-binary labels that range from -1 to 1 , where anchor and positive pairs were assigned the label 1, anchor and (hard) negative pairs the label -1 .

3.3 Generating representations

The Greek BERT and the domain-adapted Greek BERT models take a maximum input sequence of 512 tokens and generate word embeddings with a dimension of 768. Similarly, the Greek Longformer model generates word embeddings with the same dimension, with a maximum input sequence of 4,096 tokens. In order to generate document-level embeddings, a mean pooling layer is applied, to pool all the word embeddings into a single vector representation that is then normalized with L2 normalization. The resulting output constitutes the domain-adapted representations that include semantic information from contrastive learning.

4 Results and discussion

This section describes the experimental setup and the experimental results of our proposed approach to domain-adapted contrastive representation learning for clustering tasks.

4.1 Experimental setup

The Greek BERT and Longformer domain adaptation is performed in a self-supervised manner using the MLM and TSDAE training objectives, following the same setup as described in the published work where it was first presented [28, 33]. We extended the training of the LMs with contrastive learning using the triplet, contrastive and Cosine similarity loss functions and trained each pre-trained model on each source domain separately and combined, resulting in 4 models per LM. For TL, the models are evaluated using sentence triplets consisting of an anchor, a positive example and a negative example. The evaluation considers the correct outcome to be when the distance between the anchor and positive example is smaller than the distance between the anchor and negative example. The distance metric can be either Manhattan, Euclidean or Cosine, and the accuracy of each metric is calculated by dividing the correctly predicted triplets with the total number of triplets.

For CL, the models are evaluated using a dataset created from the TL dataset, where each training example is assigned a label of 1 for anchor-positive pairs and 0 for anchor-negative pairs. The evaluation involves calculating the distance between embeddings based on the assigned labels. For pairs labeled 1, the aim is to reduce the distance between embeddings, signifying similarity, while for pairs labeled 0, the goal is to increase the distance, indicating dissimilarity. The models are trained to minimize the CL loss, which is computed using a combination of squared distances and ReLU activations, as described in Eq. 4.

For CSL, the models are evaluated using a dataset derived from the TL dataset, where training examples now include non-binary labels ranging from -1 to 1 . The labeling scheme designates pairs of anchor and positive examples as 1 , and pairs of anchor and (hard) negative examples as -1 . The evaluation involves measuring the Cosine similarity between embeddings generated by the model for sentence pairs and comparing the result to the gold similarity scores (input labels).

To handle large bodies of text in documents, we conducted experiments using summarization techniques that are used in place of feeding the larger documents to the Longformer-based models. Specifically, we explored the use of LexRank [35], an unsupervised graph-based method

for text summarization that ranks the importance of sentences based on eigenvector centrality. This approach allows us to condense the potentially useful information in a document into a shorter, more manageable summary.

To evaluate the performance of our proposed approach, we conducted the experiments on the clustering dataset presented in Sect. 3.1.3 and used the Silhouette coefficient (SC) as a goodness-of-fit metric, which measures the cohesion and separation of a data point by calculating its similarity to its own cluster compared to the other clusters, respectively. An SC of 1 indicates that the data points are very well matched to their own cluster, while a value of -1 indicates that they are more similar to points that belong to another cluster.

However, this metric can be sensitive to the choice of input parameters, such as the number of clusters, and may not always provide a reliable measure of model quality in cases where the number of clusters is not well-defined. To alleviate this issue, we evaluate the within-cluster variation of the data points for different numbers of clusters, and select the number of clusters that leads to the lowest sum of squared errors. A summary of the experimental setup parameters is shown in Table 3.

The training of the domain-adapted Greek BERT (DAGB) took approximately 11 days, while pre-training the initial Greek Longformer (GLF) on the domain adaptation dataset and domain-adapting the fully pre-trained Greek Longformer (DAGLF) took approximately 6 days. The training was conducted on a computer with a 24-core Intel CPU and two Nvidia RTX A6000 graphics cards with 48GB memory each. While the difference in training time can be attributed to the use of the more efficient global attention mechanism in the Longformer architecture compared to the attention mechanism used in the BERT-based models, the domain adaptation dataset also contributed to this. In contrast, the complete pre-training of the Greek Longformer took approximately 45 and 60 days for pre-training and triplet loss, respectively [27]. The increase in triplet-loss training time is due to not leveraging any kind of parallelization.

4.2 Experimental results

Initially, the Greek Longformer pre-training was carried out with randomly initialized weights (from scratch) using the same training objective and the proposed setup, with the input sequences having a maximum length of 4,096 tokens and achieved a relatively low accuracy of 37.05% on the evaluation set using the MLM pre-training task. The fully pre-trained Greek Longformer achieved a score of 77.65% on the same task. The input sequences were tokenized and padded or truncated to a maximum of 512 tokens for Greek BERT and 4096 tokens for Greek

Table 3 Summary of experimental setup parameters

Parameter	Description
LM Training Objectives	MLM and TSDAE
Triplet-Loss Functions	Triplet, contrastive and Cosine similarity
Models	Greek BERT and Longformer
Distance Metrics	Manhattan, Euclidean, Cosine
Triplet Labeling Scheme	Binary (Anchor-Positive pairs: 1, Anchor-Negative pairs: 0) Non-binary (−1 to 1)
Clustering Goodness-of-fit metric	Silhouette Coefficient (SC)

Longformer, and the models were domain-adapted for a total of 3 epochs, achieving an accuracy of 76.16% and 75.74% on the MLM evaluation set, respectively. Pre-training using TSDAE, the Greek BERT and Greek Longformer LMs yielded an accuracy of 75.32% and 72.68%, respectively, achieving slightly lower performance than the MLM pre-training objective. Therefore, the following experiments that evaluate the performance of the domain-adapted models utilize the Domain-Adapted (DA) model variants that were adapted using MLM.

In Tables 4, 5 and 6, we present the training results of the contrastive training approaches for each pre-trained language model using the triplet loss, contrastive loss and Cosine similarity loss evaluations. We included the results of the Greek Longformer version that was randomly initialized and pre-trained on the domain adaptation datasets, denoted as TL-GLF*, CL-GLF* and CLS-GLF*, respectively. However, the proper domain-adapted versions, denoted as TL-GLF, CL-GLF and CLS-GLF, utilize the properly pre-trained Greek Longformer as the foundational LM. All LMs achieved high accuracy scores on each distance metric and dataset with our domain-adapted Greek BERT variants (TL/CL/CSL-DAGB) and triplet-loss Greek Longformer (TL-GLF) outperforming the Greek BERT variants (TL/CL/CSL-GB) model in all datasets and metrics.

In comparing two model variants, the DAGB-based models performed better than the initial Longformer-based model across all metrics. This highlights the significance of the first-phase pre-training process, which the DAGB models underwent by being initialized with Greek BERT weights. The results of the comparison provide further evidence for the importance of this initial pre-training step in developing high-performing language models since our GLF model was pre-trained on a much smaller scale than Greek BERT. This is further supported by the increase in performance of the domain-adapted Greek Longformer variants that utilize the fully pre-trained GLF (DAGLF-variants), which achieve similar results to the DAGB-variants, closing the gap between the two architectures.

In our evaluation of LMs trained using the contrastive learning approach with the contrastive loss metric

Table 4 Triple loss evaluation results of each LM trained using the contrastive learning approach. TL-GLF* denotes the initial version of the Greek Longformer

Model	Training steps (approx.)	Acc_{Cos}	Acc_{Man}	Acc_{Euc}
TL-GB				
- internet	6k	90.78	90.76	90.77
- press	8k	88.93	88.93	88.94
- social	10k	91.48	91.94	91.74
- combined	22k	90.32	90.33	90.34
TL-DAGB				
- internet	6k	97.87	97.86	97.93
- press	8k	92.39	92.39	92.44
- social	10k	98.44	98.49	98.47
- combined	22k	96.23	96.24	96.25
TL-GLF*				
- internet	21k	93.05	93.04	93.05
- press	56k	90.15	90.18	90.13
- social	19k	94.98	94.96	94.98
- combined	173k	92.30	92.31	92.29
TL-GLF				
- internet	21k	94.00	94.00	94.00
- press	56k	90.90	90.95	90.75
- social	19k	95.93	95.92	95.96
- combined	173k	92.92	92.93	92.91
TL-DAGLF				
- internet	21k	97.87	97.86	97.93
- press	56k	94.68	94.68	94.74
- social	19k	98.55	98.60	98.58
- combined	173k	96.34	96.35	96.36

(Table 5), a consistent trend can be observed. The models CL-DAGB and CL-GLF consistently outperform the CL-GB model across diverse datasets and evaluation metrics. Furthermore, the performance of the CL-GLF model, representing the initial version of the Greek Longformer, remains competitive, highlighting the efficacy of the Longformer architecture in capturing meaningful contextual information.

Table 5 Contrastive loss evaluation results of each LM trained using the contrastive learning approach. CL-GLF* denotes the initial version of the Greek Longformer

Model	Training steps (approx.)	Acc_{Cos}	Acc_{Man}	Acc_{Euc}
CL-GB				
- internet	6k	90.05	89.73	90.28
- press	8k	87.93	88.65	88.18
- social	10k	90.71	91.79	90.84
- combined	22k	89.65	89.69	90.02
CL-DAGB				
- internet	6k	96.92	97.65	97.84
- press	8k	91.61	91.95	91.69
- social	10k	97.39	97.91	97.05
- combined	22k	95.42	95.43	96.02
CL-GLF*				
- internet	21k	92.09	92.90	92.35
- press	56k	89.25	89.43	89.07
- social	19k	94.03	93.25	94.22
- combined	173k	91.15	91.05	91.63
CL-GLF				
- internet	21k	93.10	93.12	93.25
- press	56k	90.30	90.35	90.20
- social	19k	94.47	93.60	94.66
- combined	173k	91.79	91.67	92.23
CL-DAGLF				
- internet	21k	96.25	97.63	97.38
- press	56k	94.32	93.76	93.99
- social	19k	98.03	98.15	97.91
- combined	173k	95.59	95.90	95.55

In our evaluating the contrastive learning approach and with the Cosine similarity loss metric (Table 6), both CSL-DAGB and CSL-GLF consistently outperform the CSL-GB model across various datasets and evaluation metrics. Simultaneously, the CSL-GLF model, representing the initial version of the Greek Longformer, continues to demonstrate robust performance, further affirming the efficacy of the Longformer architecture in capturing and leveraging contextual information.

To evaluate the effectiveness of our proposed approach and its impact on clustering tasks, we tested the generated representations with two clustering algorithms: hierarchical agglomerative clustering (HAC) and K-means (KM). HAC is a bottom-up approach that merges document representations based on their similarity until a single cluster containing all the document representations is formed. On the other hand, KM is a top-down approach that assigns each document representation to the nearest centroid and

Table 6 Cosine similarity loss evaluation results of each LM trained using the contrastive learning approach. CSL-GLF* denotes the initial version of the Greek Longformer

Model	Training steps (approx.)	Acc_{Cos}	Acc_{Man}	Acc_{Euc}
CSL-GB				
- internet	6k	90.25	90.02	90.49
- press	8k	88.32	88.86	88.55
- social	10k	91.92	92.25	92.05
- combined	22k	90.78	90.81	91.10
CSL-DAGB				
- internet	6k	97.10	97.45	97.68
- press	8k	92.15	92.29	92.20
- social	10k	97.80	98.05	97.93
- combined	22k	96.00	96.10	96.40
CSL-GLF*				
- internet	21k	92.75	93.35	93.10
- press	56k	90.20	90.60	90.30
- social	19k	95.20	94.80	95.35
- combined	173k	92.80	92.70	93.20
CSL-GLF				
- internet	21k	93.00	93.70	93.30
- press	56k	90.80	91.20	90.60
- social	19k	95.80	95.40	95.75
- combined	173k	93.50	93.40	93.90
CSL-DAGLF				
- internet	21k	97.20	97.55	97.80
- press	56k	94.60	94.20	94.50
- social	19k	98.20	98.10	98.00
- combined	173k	96.00	96.30	96.10

updates the centroids until convergence to create non-overlapping clusters.

In the subsequent experiments evaluating the goodness-of-fit in clustering tasks, we will utilize the models from the best contrastive learning approach, which, in this case, is triplet loss. We evaluated each model using four variants of each dataset, each with different preprocessing steps. The full preprocessing (Full PP) variant included removal of users/links/email addresses, text cleanup (removing non-ASCII characters) and lower casing, while the normalized variant only removed non-ASCII characters and converted the text to lowercase. The raw variant refers to using the collected text without any preprocessing steps applied to it, while the summary variant used summarized text instead of Longformer-based models to process longer sequences.

By testing each model with these different datasets and preprocessing steps, we assess the effectiveness of preprocessing in combination with our proposed approach on clustering tasks and present the results of each model on

Table 7 Per model goodness-of-fit results based on the Silhouette coefficient on the internet dataset

Internet Models	Raw		Full PP		Normalized		Summary	
	HAC	KM	HAC	KM	HAC	KM	HAC	KM
GB	0.33	0.31	0.33	0.34	0.34	0.31	0.33	0.33
TL-GB	0.33	0.30	0.32	0.35	0.33	0.31	0.01	0.01
TL-GB-C	0.33	0.31	0.30	0.31	0.33	0.33	0.01	0.01
DAGB	0.38	0.40	0.39	0.39	0.38	0.37	0.60	0.60
TL-DAGB	0.38	0.45	0.38	0.45	0.52	0.57	0.01	0.01
TL-DAGB-C	0.43	0.45	0.51	0.52	0.50	0.54	0.01	0.01
GLF*	0.83	0.87	0.90	0.90	0.52	0.52	–	–
TL-GLF*	0.52	0.56	0.56	0.59	0.57	0.57	–	–
TL-GLF-C*	0.76	0.76	0.79	0.79	0.78	0.78	–	–
GLF	0.86	0.89	0.92	0.93	0.55	0.55	–	–
TL-GLF	0.55	0.58	0.59	0.62	0.60	0.60	–	–
TL-GLF-C	0.79	0.79	0.82	0.82	0.81	0.81	–	–

the text from the internet, social and press source domains in Tables 7, 8 and 9, respectively.

The results indicate that the combined use of BERT and Longformer-based models outperform the pure BERT-based variants in all three source domains in the document clustering task. Our approach leverages the representation generation capabilities of the domain-adapted Greek BERT and the Greek Longformer by encoding the sequences that are up to 512 tokens with the former and sequences greater than 512 tokens with the latter. The GLF, TL-GLF and TL-GLF-C models are combined with the best performing DAGB-based models on each domain, where at least one variant always outperforms the best Greek BERT (GB) variant.

On the internet source dataset, the triplet-loss domain-adapted Greek BERT trained on all datasets (TL-DAGB-C) as presented in Sect. 3.2.2 in combination with the fully pre-trained Greek Longformer (GLF) on a fully pre-processed text, achieved the best results with an SC of 0.90, indicating a close to perfect goodness-of-fit. Similarly, on the social source dataset, the triplet-loss domain-adapted Greek BERT (TL-DAGB) in combination with the triplet-loss Greek Longformer (TL-GLF), both trained on the domain-specific dataset, outperformed the rest with an SC of 0.86. A similar pattern can be observed in the press source domain, where the TL-DAGB model in combination with the Greek Longformer (GLF) model achieves the best score with an SC of 0.85.

The overall results show that both the domain adaptation as well as the contrastive learning approach to further train the transformer-based LMs improve the results over the representations gained from the models they are based on. While the use of the Greek Longformer with a triplet-loss BERT-based model performed better in two datasets, internet and press, the use of the triplet-loss Greek

Longformer performed significantly better compared to the rest on the same social source dataset. This potentially can be attributed to the use of non-formal language where the communication norms are often ignored, and further training of the LM may be able to learn.

4.3 Limitations and challenges

Our approach heavily relies on domain-specific data for pre-training and evaluation. While we achieved promising results within the selected domains, the effectiveness of the models may vary across different domains. Generalizing the performance of our models to unseen domains requires further exploration and validation. Additionally, the applicability of our approach to niche or specialized domains may be limited, as the pre-trained models may not capture domain-specific characteristics effectively.

Moreover, training and fine-tuning large transformer-based models like BERT and Longformer entail significant computational resources and time. The computational cost associated with our approach limits its scalability and accessibility, particularly for settings with constrained computational resources. Furthermore, the extensive training time may hinder rapid experimentation or deployment of our models in time-sensitive applications.

Finally, the performance of contrastive learning methods employed in our approach can be sensitive to hyperparameters such as learning rates, batch sizes and margin values. While we conducted experiments to optimize these hyperparameters, robust hyperparameter tuning strategies are necessary to ensure consistent performance across different datasets and tasks.

Table 8 Per model goodness-of-fit results based on the Silhouette coefficient on the social dataset

Social Models	Raw		Full PP		Normalized		Summary	
	HAC	KM	HAC	KM	HAC	KM	HAC	KM
GB	0.34	0.12	0.36	0.38	0.32	0.34	0.11	0.13
TL-GB	0.33	0.34	0.35	0.36	0.34	0.31	0.01	0.01
TL-GB-C	0.36	0.37	0.38	0.38	0.36	0.35	0.01	0.01
DAGB	0.34	0.13	0.31	0.28	0.34	0.31	0.70	0.70
TL-DAGB	0.39	0.41	0.43	0.49	0.36	0.39	0.01	0.01
TL-DAGB-C	0.46	0.45	0.47	0.47	0.52	0.44	0.01	0.01
GLF*	0.33	0.43	0.47	0.52	0.48	0.51	–	–
TL-GLF*	0.78	0.80	0.84	0.86	0.83	0.84	–	–
TL-GLF-C*	0.75	0.77	0.82	0.83	0.81	0.81	–	–
GLF	0.36	0.46	0.50	0.55	0.52	0.54	–	–
TL-GLF	0.80	0.82	0.86	0.89	0.86	0.87	–	–
TL-GLF-C	0.77	0.79	0.84	0.85	0.83	0.83	–	–

Table 9 Per model goodness-of-fit results based on the Silhouette coefficient on the press dataset

Press Models	Raw		Full PP		Normalized		Summary	
	HAC	KM	HAC	KM	HAC	KM	HAC	KM
GB	0.34	0.13	0.42	0.44	0.39	0.42	0.34	0.13
TL-GB	0.33	0.31	0.34	0.34	0.33	0.34	0.01	0.01
TL-GB-C	0.33	0.31	0.40	0.40	0.38	0.40	0.01	0.01
DAGB	0.34	0.30	0.33	0.31	0.45	0.38	0.11	0.05
TL-DAGB	0.53	0.53	0.49	0.55	0.45	0.49	0.01	0.01
TL-DAGB-C	0.54	0.58	0.51	0.51	0.46	0.50	0.01	0.01
GLF*	0.67	0.67	0.83	0.85	0.49	0.60	–	–
TL-GLF*	0.61	0.61	0.63	0.62	0.61	0.59	–	–
TL-GLF-C*	0.72	0.74	0.76	0.76	0.74	0.73	–	–
GLF	0.69	0.69	0.86	0.88	0.51	0.62	–	–
TL-GLF	0.63	0.63	0.65	0.64	0.63	0.61	–	–
TL-GLF-C	0.74	0.76	0.78	0.78	0.76	0.75	–	–

5 Conclusions

In this paper, we propose a domain-adapted contrastive learning approach to low-resource language representations for document clustering tasks and evaluating difference contrastive loss functions on manually generated datasets. Our study initially investigated the effectiveness of the pre-training task for adapting language models to specific domains and combining domain-adapted Greek BERT and Greek Longformer models using contrastive learning across three source domains, further fine-tuning them. Our findings demonstrated that both domain adaptation and contrastive learning approaches using triplet-loss training improved the performance of transformer-based language models for document clustering tasks.

We demonstrate notable accuracy improvements, with our proposed approach consistently outperforming baseline methods across various datasets and evaluation metrics.

These results underscore the significance of leveraging domain-specific data and advanced pre-training techniques for enhancing the performance of clustering tasks in low-resource language settings. Additionally, our experiments highlight the efficacy of combining language models optimized for different sequence lengths and the importance of effective document representations in unsupervised learning tasks.

Future work could further explore the effectiveness of different pre-training strategies for domain-adapting language models and combining weights from other pre-trained models. Finally, considering the performance that was achieved in our experimental setting, another direction for future work is to explore more approaches to automatically create large contrastive learning datasets from a smaller annotated corpus.

Future work could further explore the effectiveness of different pre-training strategies for domain-adapting language models and combining weights from other pre-

trained models. Additionally, a more comprehensive comparison of text representation methods, including traditional methods, could be explored to better understand their effectiveness in our experimental context. Finally, considering the performance that was achieved in our experimental setting, another direction for future work is to explore more approaches to automatically create large contrastive learning datasets from a smaller annotated corpus.

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Data availability The data supporting the findings of this study are accessible through the corresponding author, albeit subject to certain restrictions. These data were used under license for the current study and are not publicly available at present. However, the data are available from the corresponding author upon reasonable request. The release of these data is contingent upon obtaining permission from project partners and will be made publicly available subsequent to obtaining necessary clearances at <https://huggingface.co/dimitriz>.

Code Availability The pre-trained language models are available at <https://huggingface.co/dimitriz>. The code supporting the findings of this study will be made available upon publication of this work at <https://github.com/dimitriz>.

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